

SIMATS ENGINEERING ASSESSMENT TOOL 2

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 50%

QUESTIONS: 2

Duration : 1 Hour
Total Marks:30

Annexure A

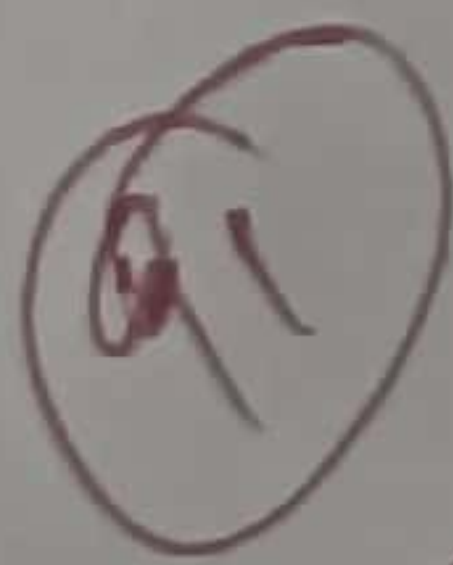
Constraint-Based Problem Solving

Code of Conduct:

I G. Pallavi (192311208) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

G. Pallavi


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1. A college must assign 4 employees (E1, E2, E3, E4) to 4 administrative tasks (Admissions, Accounts, Records, IT Support) such that:

Constraints:

- i. E1 can only handle Admissions or Accounts
- ii. E2 must be assigned before E3 (Admissions < Accounts < Records < IT Support)
- iii. Only one employee per task
- iv. E4 cannot handle Records
- v. All employees must be assigned exactly one task

Apply Backtracking Search to find a valid assignment with all intermediate steps and constraint checks. State the final valid schedule.

2. A security patrol robot operates in a 6x6 building-floor grid from Start (S) to a checkpoint (G). Certain zones are restricted (no-entry) and some corridors carry extra monitoring cost.

Constraints:

- i. Movement allowed: Up, Down, Left, Right
- ii. Normal move cost = 1, monitored-corridor move cost = 2
- iii. Cannot pass through restricted (no-entry) cells
- iv. The robot must minimize total path cost

Using the above constraints and Uniform Cost Search (UCS), show step-by-step node expansion and priority-queue updates, and determine the optimal path and cost.

Employee - Task Assignment Using Backtracking Search:-

Given Employees:

E_1, E_2, E_3, E_4

Tasks:

1. Admissions 2. Accounts 3. Records 4. IT Support

Task order:-

Admissions < Account < Records < IT Support

Constraints:-

- C1: $E_1 \rightarrow$ Admissions or Accounts only.
- C2: E_2 's task must come before E_3 's task
- C3: One employee per task.
- C4: $E_4 \neq$ Records
- C5: Every employee gets exactly one task.

Step 1: Start Backtracking:

E_1 can only be assigned to Admissions or Accounts.

Try:

$E_1 \rightarrow$ Admissions

Constraint Check:

$\rightarrow E_1$ is allowed

$\rightarrow$ Admissions is currently unassigned.

Current Assignment:

$E_1 \rightarrow$ Admissions

Step 2: Assign E_2

The remaining tasks are:

- Accounts
- Records
- IT Support

Try:

$E_2 \rightarrow$ Accounts

Constraint check:

- Accounts is available

- E_2 is assigned to a valid task.

Current task:

• $E_1 \rightarrow$ Admissions

• $E_2 \rightarrow$ Accounts

Step 3: Assign E_3

The constraint states that E_2 must be assigned before E_3 .

Since: $E_2 \rightarrow$ Accounts

$\rightarrow E_3$ must be assigned to a task after accounts.

The available tasks after Accounts are:

- Records
- IT Support

Try:

$E_3 \rightarrow$ Records

Constraint check:

- Records is available.

- Accounts < Records.

- Therefore, E_2 is assigned before E_3 .

Assignment :

- $E_1 \rightarrow$ Admissions
- $E_2 \rightarrow$ Accounts
- $E_3 \rightarrow$ Records

Step 4 : Assign E_4

The only remaining task is :

IT Support

Try :

$E_4 \rightarrow$ IT Support

Constraint Checks :

- $\rightarrow$ IT Support is available
- $\rightarrow E_4$ is not assigned to Records.
- $\rightarrow$ All four employees now have exactly one task.

Backtracking Search Tree :-

Start



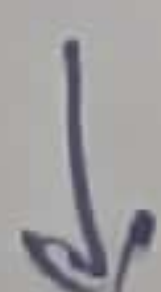
$E_1 \rightarrow$ Admissions



$E_2 \rightarrow$ Accounts



$E_3 \rightarrow$ Records



$E_4 \rightarrow$ IT Support

is required because the first valid

$\Rightarrow$ No backtracking
choices satisfy all constraints.

Constraint Verification:-

Constraint

1. E_1 Can only handle Admissions or Accounts
2. E_2 must be before E_3
3. Only one employee per task.
4. E_4 Cannot Handle Records.
5. All employees assigned exactly one task

Verification

$E_1 \rightarrow$ Admissions

Accounts < Records

Each task has one employee.

$E_4 \rightarrow$ IT Support

E_1, E_2, E_3, E_4 all assigned.

2. Security Patrol Robot Using Uniform Cost Search:

Given:

- Grid Size : 6×6
- Movement : Up, Down, Left, Right
- Normal movement cost = 1
- Monitored corridor cost = 2.
- Restricted cells cannot be entered.
- UCS selects the node with the lowest cumulative path cost.

UCS formula:

$g(n)$ = total cost from S to n.

5 Steps:-

1. Start from S with cost 0.
2. Expand the available neighbouring cells.
3. Ignore restricted cells.
4. Add 1 for a normal movement and 2 for a monitored movement.
5. Insert the new nodes into the priority queue according to their cumulative cost.
6. Always expand the node with the smallest cost.
7. Continue until G is reached.

Node Expansion and priority Queue :

<u>Step</u>	<u>Expanded Node</u>	<u>Cost</u>	<u>Priority Queue</u>
1.	S	0	$(1,2):1, (2,1):1$
2.	$(1,2)$	1	$(2,1):1, (1,3):2$
3.	$(2,1)$	1	$(1,3):2, (3,1):2$
4.	$(1,3)$	2	$(3,1):2, (2,3):4$
5.	$(3,1)$	2	$(3,2):3, (2,3):4$
6.	$(3,2)$	3	$(4,2):4, (3,3):5$
7.	$(4,2)$	4	$(2,3):4, (5,2):5$
8.	$(2,3)$	4	$(5,2):5, (3,3):5$
9.	$(5,2)$	5	$(3,3):5, (5,3):6$

$\Rightarrow$ Restricted cells are discarded whenever encountered, and monitored cells add cost 2.

Therefore :

UCS always expands the node having the minimum cumulative cost. Therefore, when the goal G is selected, the resulting route is the optimal minimum-cost path.